DATA SCIENCE MINOR

PROJECT REPORT

(Project Semester August-December 2026)

PRODUCTPULSE: AN INTERACTIVE DATA SCIENCE FRAMEWORK FOR E-COMMERCE CUSTOMER CHURN PREDICTION, BEHAVIOURAL COHORT SEGMENTATION, AND REVENUE RETENTION OPTIMIZATION

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DECLARATION

I, Arayana Sood, student of B.Tech. Computer Science and Engineering under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

The analysis, implementation details and results presented in this report are based on the project work and the public dataset described herein. Sources used for background information and methodology have been acknowledged in the references. Assumptions used in the economic illustration are explicitly identified as assumptions and are not presented as observed company results.

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CERTIFICATE

This is to certify that Arayana Sood bearing Registration no. 12412379 has completed INT345 (Data Science Minor) project titled, **"PRODUCTPULSE: AN INTERACTIVE DATA SCIENCE FRAMEWORK FOR E-COMMERCE CUSTOMER CHURN PREDICTION, BEHAVIOURAL COHORT SEGMENTATION, AND REVENUE RETENTION OPTIMIZATION"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

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Finally, I thank my family and peers for their continuous support and encouragement during the completion of ProductPulse.

Arayana Sood

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ABSTRACT

ProductPulse is an interactive data science framework designed to analyse e-commerce customer behaviour and identify users who may become inactive over a short prediction horizon. The project uses event-level behavioural data containing product views, cart additions, cart removals, purchases, prices, user identifiers and session identifiers. The data is transformed into user-level behavioural features and used for three complementary tasks: short-horizon retention classification, behavioural cohort segmentation and low-dimensional visual analysis.

Four supervised classification approaches are evaluated: Logistic Regression, Decision Tree, Random Forest and Linear Support Vector Machine (SVM). Because the target classes are imbalanced, model selection considers precision, recall, F1-score and ROC-AUC rather than accuracy alone. Linear SVM produces the highest F1-score (0.3586) among the compared models, with a ROC-AUC (0.7430) that is effectively tied with the other models. K-Means clustering identifies two behavioural groups, and Principal Component Analysis (PCA) provides a two-dimensional representation explaining 54.21% of the variance.

The report also presents an illustrative retention-value calculation. Average order value (AOV), customer acquisition cost (CAC) and recovery-rate scenarios are treated as hypothetical assumptions; they are not measured financial outcomes from the dataset or from a real company. ProductPulse is intended as a decision-support prototype, not as proof that a particular intervention will causally recover customers.

Keywords: e-commerce analytics, customer inactivity, churn prediction, classification, Linear SVM, K-Means, PCA, customer retention.

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LIST OF ABBREVIATIONS

Table 1: Abbreviations used in this report

Abbreviation	Meaning
AOV	Average order value
API	Application programming interface
AUC	Area under the curve
CAC	Customer acquisition cost
CART	Classification and regression trees (Decision Tree)
CSV	Comma-separated values
EDA	Exploratory data analysis
EVR	Explained variance ratio
GMV	Gross merchandise value
PCA	Principal component analysis
ROC	Receiver operating characteristic
SVM	Support vector machine
UTC	Coordinated universal time
WCSS	Within-cluster sum of squares

1. INTRODUCTION

1.1 Background

E-commerce platforms record large volumes of behavioural events as customers browse products, add items to carts, remove items and complete purchases. These interactions provide signals about customer engagement and purchase intent. Analysing such signals can help an online retailer identify users whose recent activity suggests a higher risk of becoming inactive.

Retention is widely regarded as economically attractive for online retailers, because returning customers tend to purchase repeatedly and do not require the repeated acquisition spending that new customers do [1]. Switching between competing online stores costs the customer very little, and rising advertising costs and stricter data-privacy rules make customer acquisition progressively more expensive. Retailers therefore benefit from methods that detect early behavioural signs of disengagement while there is still an opportunity to respond.

Customer churn is often used to describe the loss of customers over a defined period. In this project the target is operationalised more narrowly: a user is treated as retained if at least one interaction is observed in the defined target window, and as churned or at-risk if no interaction is observed in that window. The model therefore predicts short-horizon inactivity under the project's chosen time-window definition; it does not establish permanent customer loss.

1.2 Problem Statement

The project addresses the challenge of converting raw e-commerce event logs into actionable user-level signals. The raw data is event-oriented, while retention decisions require a user-level view of behaviour. The work therefore requires temporal aggregation, feature engineering, classification under class imbalance, behavioural segmentation, and an interface through which a user can inspect predictions and analytical summaries.

1.3 Objectives

- Prepare and aggregate event-level e-commerce data into a user-level analytical dataset with a leakage-free temporal design.
- Explore customer behaviour through descriptive statistics and conversion-funnel analysis.
- Build and compare supervised models for short-horizon retention (inactivity) classification, with emphasis on metrics suited to imbalanced classes.

- Segment users into behavioural cohorts using K-Means clustering.
- Apply PCA to visualise the dominant structure in the behavioural feature space.
- Provide a sensitivity (what-if) analysis that shows how changes in behaviour move a user's decision score across the SVM boundary.
- Illustrate how retention scenarios can be translated into potential economic value using clearly stated assumptions.
- Present the analytical outputs through the ProductPulse application as a decision-support prototype.

1.4 Scope and Limitations

The project focuses on one monthly event file (December 2019) from a public cosmetics e-commerce dataset [2] and on the time windows defined in the project methodology. Its conclusions are limited by the coverage of the dataset, the chosen target definition, class imbalance, and the absence of verified company-specific financial inputs. A prediction is a statistical estimate, not a guarantee of future behaviour. Scenario simulations are not causal estimates of intervention impact. Further limitations are discussed in Section 5.3.

1.5 System Architecture

ProductPulse is organised as a layered system in which a Python analytics pipeline produces the models and summaries, a FastAPI service exposes them, and a React/TypeScript interface presents them to the user. Figure 1 shows the end-to-end workflow and Table 2 summarises the responsibility of each layer.

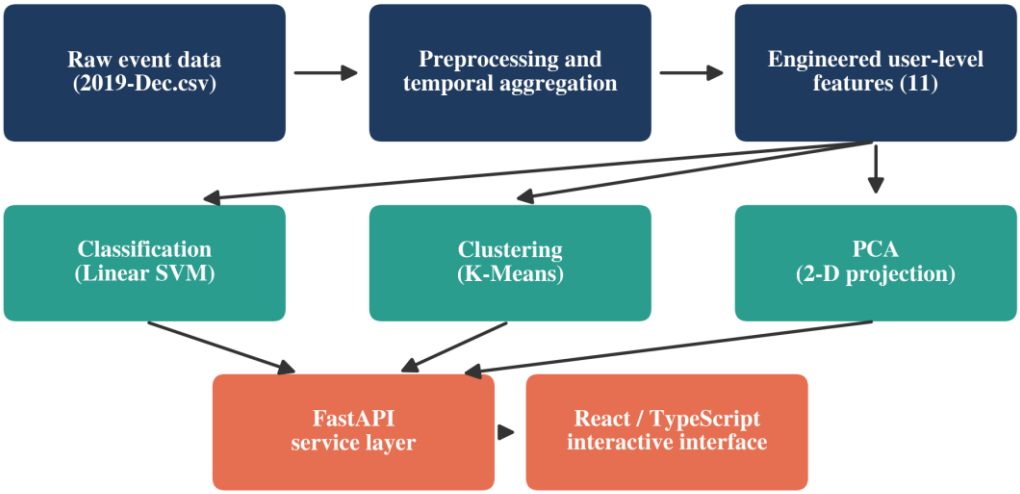


Figure 1: High-level ProductPulse system workflow

Table 2: Architecture layers of ProductPulse

Layer	Responsibility	Technology / component
Data layer	Load and prepare event-level records; aggregate to user-level features.	CSV dataset; Python (Pandas, NumPy)
Analytics layer	Train and evaluate classification, clustering and PCA analyses.	scikit-learn [3]
API layer	Expose predictions, scenario evaluations and analytical outputs to the interface.	FastAPI
Presentation layer	Display dashboard, individual analysis, segments, performance and insights.	React / TypeScript

1.6 Report Organisation

Chapter 2 describes the dataset source, temporal design and feature engineering. Chapter 3 presents the exploratory data analysis. Chapter 4 reports five analyses (classification, clustering, PCA, sensitivity analysis and retention economics), each organised into introduction, general description, requirements and formulas, results, visualisation and conclusion. Chapter 5 concludes the report and describes the ProductPulse application, Chapter 6 outlines future scope and Chapter 7 lists the references.

2. SOURCE OF DATASET

2.1 Dataset Source

The project uses the December 2019 event file, 2019-Dec.csv, from the public dataset "eCommerce Events History in Cosmetics Shop", published on Kaggle by Michael Kechinov and collected from an online cosmetics store served by the REES46 marketing platform [2]. The dataset contains timestamped product-interaction events. The raw December file used in the project contains 3,533,286 records and 9 columns.

The dataset documentation describes event records such as product views, cart additions, cart removals and purchases, together with product, category, brand, price, user and session information. Digital cosmetics retail is a suitable domain for behavioural analytics because customers often browse several products, add and remove items from the cart, and repeat purchases over time.

2.2 Raw Dataset Structure

Each record in the raw file describes a single user action. Table 3 lists the nine fields.

Table 3: Raw dataset fields and descriptions

Field	Description
event_time	Timestamp at which the event occurred (UTC).
event_type	Type of interaction: view, cart, remove_from_cart or purchase.
product_id	Identifier of the product involved in the event.
category_id	Identifier of the product category.
category_code	Hierarchical or textual category code, where available.
brand	Brand associated with the product, where available.
price	Recorded product price.
user_id	Identifier of the user who generated the event.
user_session	Identifier of the user session.

The four event types form a natural behavioural hierarchy: a **view** reflects browsing and discovery, a **cart** event reflects purchase intent, a **remove_from_cart** event reflects hesitation or friction, and a **purchase** event is the final conversion.

2.3 Temporal Definition and Target

To avoid data leakage and look-ahead bias, December 2019 is divided into two non-overlapping windows. The observation window T_obs covers 1–21 December 2019 (00:00:00 to 23:59:59 UTC) and the target window T_target covers 22–31 December 2019. Behavioural features are derived only from the observation window; the target label is derived only from the target window, as shown in Table 4.

Table 4: Target class definition

Target class	Operational definition	Interpretation
Retained (y = 1)	At least one event in 22–31 December 2019.	User returned or remained active in the target window.
Churned / at-risk (y = 0)	No event in 22–31 December 2019.	User was inactive during the target window.

This is a short-horizon inactivity label. It should not be interpreted as verified account cancellation or permanent churn.

2.4 User-Level Dataset

Aggregating the observation-window events by user and attaching the target label yields a labelled analytical cohort of 273,366 users. This cohort is the basis of all analyses in the following chapters.

2.5 Data Preparation

Data preparation follows the steps below. The pipeline is implemented in Python with Pandas, NumPy and scikit-learn [3].

1. Parse event_time as a datetime field and split the records into the observation and target windows.
2. Use user_id as the aggregation key to build a user-level table.
3. Count event types, sessions and active days for each user.
4. Derive engagement and conversion ratios from the aggregated counts, using a +1 term in ratio denominators so that users with zero views do not cause division by zero.
5. Create the binary target label from activity in the target window; keep all target-window information out of the predictor features.

6. Standardise the numeric features before distance- and margin-based methods (SVM, K-Means, PCA) are applied, as given in Eq. (1).

$$z_j = \frac{x_j - \mu_j}{\sigma_j} \tag{1}$$

where x_j is the raw value of feature j, μ_j its mean and σ_j its standard deviation.

2.6 Feature Engineering

Eleven fixed-dimensional behavioural features are engineered for every user from the observation window. They are listed in Table 5. Four of them are derived quantities defined in Eq. (2) to Eq. (5).

Table 5: User-level behavioural features

Feature	Definition
total_views	Number of product-view events.
total_carts	Number of cart-addition events.
total_purchases	Number of purchase events.
total_removes	Number of remove_from_cart events.
total_sessions	Number of distinct user sessions.
active_days	Number of distinct calendar days with activity.
activity_duration_days	Span between first and last event, in days.
total_events	Total number of recorded events for the user.
events_per_session	Average number of events per session.
purchase_view_ratio	Purchases relative to views (smoothed).
cart_view_ratio	Cart additions relative to views (smoothed).

$$\text{events_per_session} = \frac{\text{total_events}}{\text{total_sessions}} \tag{2}$$

$$\text{purchase_view_ratio} = \frac{\text{total_purchases}}{\text{total_views} + 1} \tag{3}$$

$$\text{cart_view_ratio} = \frac{\text{total_carts}}{\text{total_views} + 1} \tag{4}$$

$$\text{activity_duration_days} = \frac{\max(t) - \min(t)}{86400} \tag{5}$$

Several features are related by construction. For example, total_events is the sum of the four event counts, and events_per_session and the two ratios are computed from other features. This redundancy is expected and is taken into account when SVM coefficients are interpreted in Section 4.1.

2.7 Summary of the Analytical Dataset

Table 6 summarises the main characteristics of the data used in the analyses.

Table 6: Summary of the analytical dataset

Item	Value
Source file	2019-Dec.csv (eCommerce Events History in Cosmetics Shop)
Raw records / columns	3,533,286 records; 9 columns
Observation window	1–21 December 2019
Target window	22–31 December 2019
Labelled users	273,366
Retained / churned (at-risk)	25,113 (9.19%) / 248,253 (90.81%)
Engineered features	11 user-level behavioural features
Train / test split	80% / 20% (N_test = 54,674)

3. EXPLORATORY DATA ANALYSIS (EDA)

3.1 Purpose of EDA

Exploratory data analysis is used to understand the distribution of user activity, identify skewed behaviour, examine the purchase funnel and assess the target-class balance before modelling. The analysis covers the N = 273,366 users in the labelled cohort.

3.2 Interaction Volume and Descriptive Statistics

Table 7: Descriptive statistics of user activity

Metric	Mean	Median	Maximum
Total views	4.71	1	482
Total cart additions	2.58	0	390
Total purchases	0.60	0	165
Total sessions	2.24	1	148
Active days	1.39	1	21

The mean values are higher than the medians for every event metric (Figure 2), indicating right-skewed distributions: a relatively small number of users generate substantially more activity than the typical user. The median purchase count of zero shows that at least half of the users in the cohort made no purchase during the observation period, and the median of one active day shows that the typical user visited on a single day only.

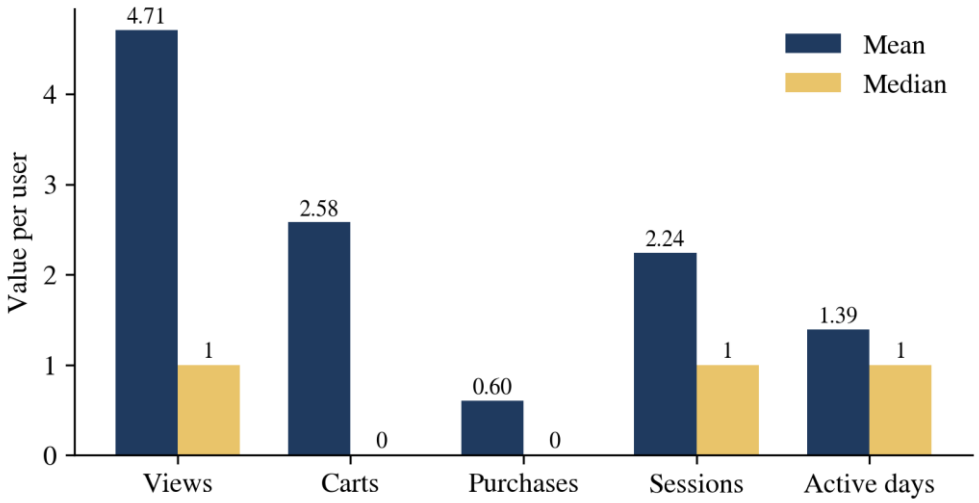


Figure 2: Mean versus median of user-level behavioural metrics

3.3 Conversion Funnel and Drop-Off

Table 8: Conversion funnel summary

Stage	Relative share	Drop-off
Product views	100.0%	—
Cart additions	15.2%	84.8% from views
Purchases	6.8%	55.3% from cart additions

The funnel in Figure 3 shows a substantial reduction between product views and later purchase events: only 15.2% of the view-stage base proceeds to a cart addition, and 6.8% to a purchase. The cart-to-purchase drop-off is 55.3%, which is consistent with $1 - 6.8/15.2$. These are event-derived funnel proportions and should not be read as a cohort conversion rate unless the same denominator is used.

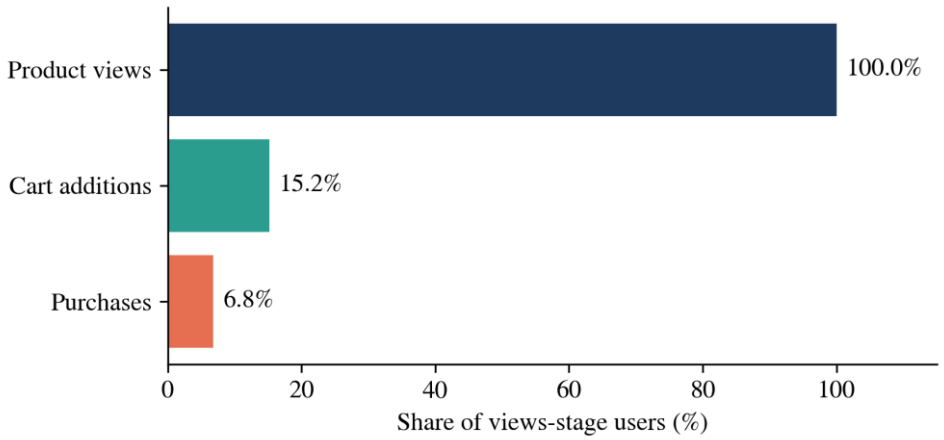


Figure 3: E-commerce conversion funnel, indexed to product views

3.4 Class Distribution

Table 9: Distribution of the target classes

Class	Users	Percentage
Churned / at-risk (y = 0)	248,253	90.81%
Retained (y = 1)	25,113	9.19%
Total	**273,366**	**100.00%**

The target is highly imbalanced, with approximately ten at-risk users for every retained user (Figure 4). A model can achieve high accuracy by favouring the majority class while failing to identify retained users. The evaluation therefore emphasises precision, recall, F1-score and ROC-AUC alongside accuracy.

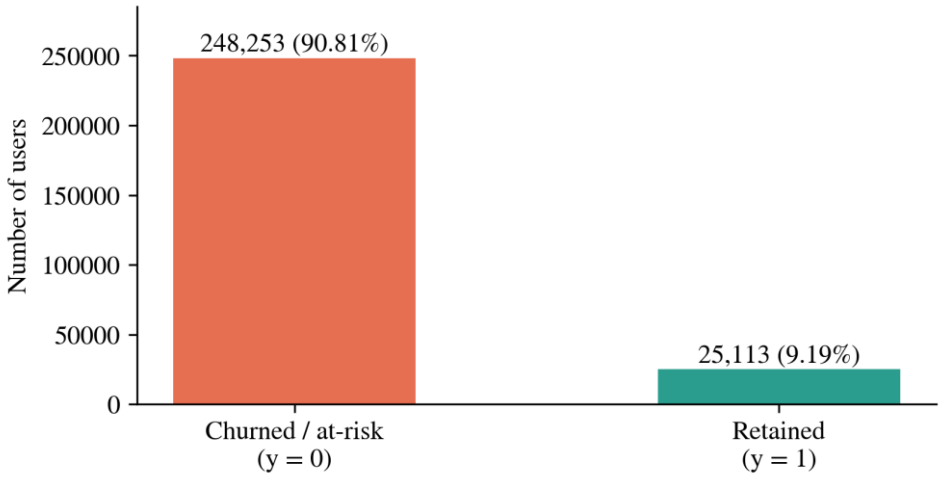


Figure 4: Class distribution of the retention target

3.5 Feature Relationships

Because many features are counts of related activity, they are positively associated: users with more sessions tend to have more active days, views and events. Ratio features are bounded and behave differently from the counts. Correlated predictors do not harm the predictive performance of a regularised linear model, but they make individual coefficients harder to interpret, because the effect of one feature can be shared with or offset by another. For this reason the SVM contributions in Section 4.1 are interpreted as a pattern rather than as isolated effects.

3.6 EDA Interpretation

- User behaviour is heterogeneous: most users show limited activity, while a smaller group is highly active.
- Purchase counts are sparse, so purchase-related features are potentially informative but unevenly distributed.
- The class imbalance makes accuracy alone an inadequate model-selection criterion.
- The funnel suggests opportunities for deeper analysis of the transition from browsing to carting and purchasing.

4. ANALYSIS ON DATASET

This chapter reports five analyses built on the user-level dataset of Chapter 2: supervised classification (Section 4.1), K-Means segmentation (Section 4.2), PCA (Section 4.3), sensitivity analysis (Section 4.4) and retention economics (Section 4.5). Each analysis follows the same structure: introduction, general description, specific requirements, functions and formulas, analysis results, visualisation and conclusion. The notation used in the formulas is summarised in Table 10.

Table 10: Notation used in the formulas of Chapter 4

Symbol	Meaning
x, x_j	Feature vector of a user and its j-th component (before standardisation).
z_j	Standardised value of feature j.
μ_j, σ_j	Mean and standard deviation of feature j.
y_i, ŷ	True and predicted class label (1 = retained, 0 = churned / at-risk).
w, b	Weight vector and bias of the Linear SVM.
C, ξ_i	SVM regularisation constant and slack variable of observation i.
f(x)	SVM decision function (signed distance from the hyperplane, up to scale).
K, S_k, μ_k	Number of clusters, k-th cluster and its centroid.
a(i), b(i)	Mean intra-cluster and nearest-cluster distance used in the silhouette score.
Σ, λ_i, v_i	Covariance matrix, i-th eigenvalue and eigenvector.
N_at-risk, L	Number of at-risk users and assumed recovery rate (percent).
AOV, CAC	Average order value and customer acquisition cost (assumed).

4.1 Supervised Classification: Short-Horizon Retention Prediction

4.1.1 Introduction

The classification task estimates whether a user will generate at least one event in the target window, based on behavioural information from the observation window. Four models from different algorithm families are compared: Logistic Regression, Decision Tree (CART), Random Forest and Linear Support Vector Machine.

4.1.2 General Description

The user-level feature table is split into a training subset (80%) and a held-out test subset (20%, N_test = 54,674). The models learn a relationship between observed behaviour and the binary target label, and all reported metrics are computed on the held-out data. The positive class is

"retained" (y = 1), so precision and recall describe how well the model identifies users who will return.

Logistic Regression is a linear probabilistic classifier; the Decision Tree partitions the feature space with axis-aligned splits; the Random Forest averages many decorrelated trees; and the Linear SVM [4], [5] finds the separating hyperplane with the largest margin. The SVM output is a signed distance from the hyperplane, not a calibrated probability.

4.1.3 Specific Requirements, Functions and Formulas

The models are implemented with the corresponding scikit-learn estimators [3] on standardised features (Eq. (1)). Because of the class imbalance, performance is judged with the following measures, where TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{7}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{8}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

ROC-AUC measures the ability of a model to rank retained users above churned users across all classification thresholds. The soft-margin Linear SVM solves the optimisation problem in Eq. (10) and classifies a user by the sign of the decision function in Eq. (11), where w is the weight vector, b the bias, C the regularisation constant and ξ_i the slack variables.

$$\min_{w,b,\xi} \tfrac{1}{2}\|w\|^2 + C\sum_{i=1}^{N}\xi_i \quad \text{s.t.} \quad y_i\,(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \tag{10}$$

$$f(x) = w^T x + b, \qquad \hat{y} = \text{sign}\,(f(x)) \tag{11}$$

4.1.4 Analysis Results

Table 11: Classification performance on the held-out test set

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.9119	0.6113	0.1121	0.1894	0.7429
Decision Tree	0.7733	0.2235	0.5935	0.3248	0.7404
Random Forest	0.8001	0.2415	0.5495	0.3355	0.7389
Linear SVM	0.8391	0.2829	0.4895	0.3586	0.7430

Linear SVM achieves the highest F1-score (0.3586) among the four models. Logistic Regression has the highest accuracy (0.9119) and precision (0.6113), but its recall is only 0.1121, so it identifies a small fraction of the retained users; its high accuracy largely reflects the majority class. The Decision Tree and Random Forest have higher recall than the SVM (0.5935 and 0.5495) but lower precision and lower F1-score.

The ROC-AUC values of all four models lie between 0.7389 and 0.7430. The SVM's value (0.7430) exceeds that of Logistic Regression (0.7429) by only 0.0001, which is not a meaningful difference; the models can be regarded as equivalent in ranking ability. Linear SVM is therefore selected on the basis of the best F1-score and its balance between precision and recall, not on accuracy or on a ROC-AUC advantage. Its precision of 0.2829 also means that many users predicted as retained may not actually return, which should be communicated wherever the output is used.

Table 12 lists the contributions reported for the Linear SVM. Active days (+1.42), activity duration (+1.15) and total purchases (+0.88) push a user towards the retained class, whereas cart removals (−0.65) and total views (−0.34) push towards the at-risk class. The negative weight on total_views should not be read as "viewing causes churn": views are strongly related to other count features (Section 3.5), so the coefficient partly adjusts for the other variables. These values describe associations in the fitted model and do not establish causality.

Table 12: Reported Linear SVM feature contributions

Feature	Contribution	Direction
active_days	+1.42	Towards retained
activity_duration_days	+1.15	Towards retained
total_purchases	+0.88	Towards retained
total_removes	−0.65	Towards at-risk
total_views	−0.34	Towards at-risk

To make the operating point of the selected model concrete, Table 13 gives the confusion matrix implied by the reported precision, recall and class prevalence. The counts are reconstructed approximately from the reported metrics (about 5,023 retained and 49,651 churned users in the test set) and should be replaced by the exact matrix from the model output if it is available.

Table 13: Approximate confusion matrix of the Linear SVM implied by the reported metrics

	Predicted retained	Predicted at-risk
Actually retained (≈ 5,023)	≈ 2,459 (TP)	≈ 2,564 (FN)
Actually churned (≈ 49,651)	≈ 6,233 (FP)	≈ 43,418 (TN)

Under this operating point the model flags about 8,700 users as likely to be retained, of whom roughly 2,460 actually return, while it misses about half of the users who do return. This illustrates why precision and recall must be read together and why the output should support prioritisation rather than automatic decisions.

4.1.5 Visualization

Figure 5 and Figure 6 compare the models by F1-score and ROC-AUC, Figure 7 shows the precision–recall trade-off behind these scores, and Figure 8 shows the reported SVM contributions.

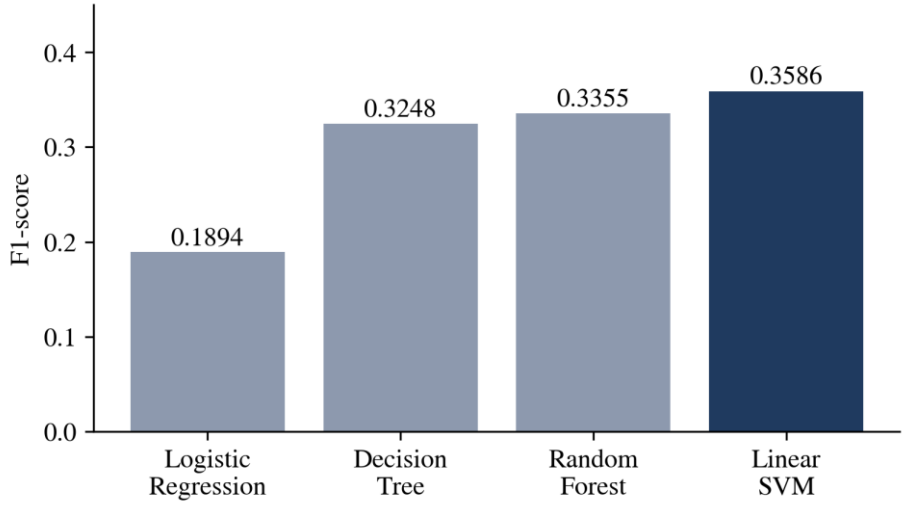


Figure 5: Model comparison by F1-score

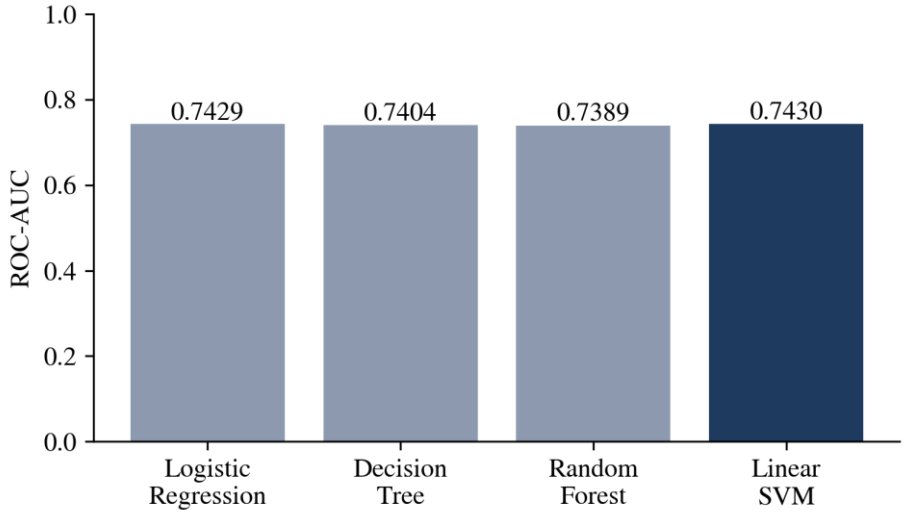


Figure 6: Model comparison by ROC-AUC

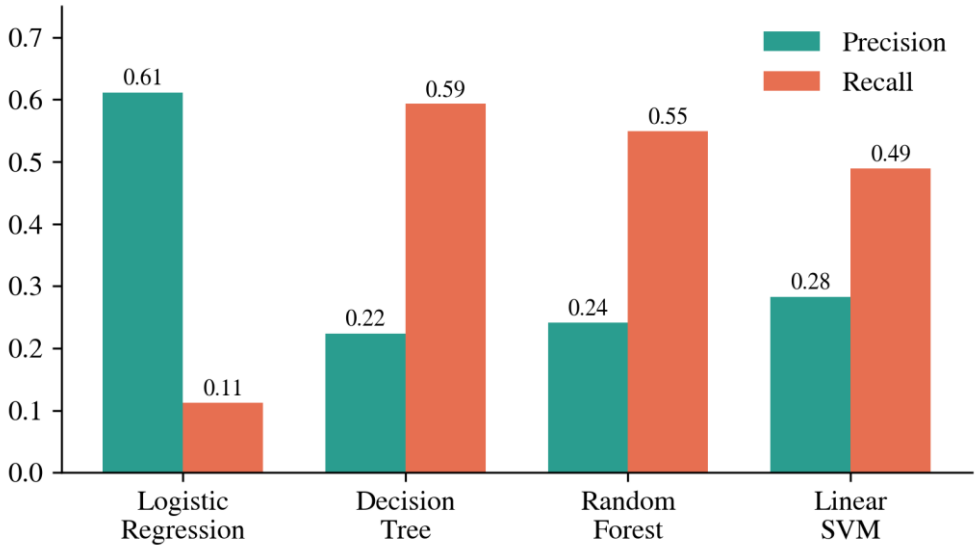


Figure 7: Precision and recall of the four classifiers

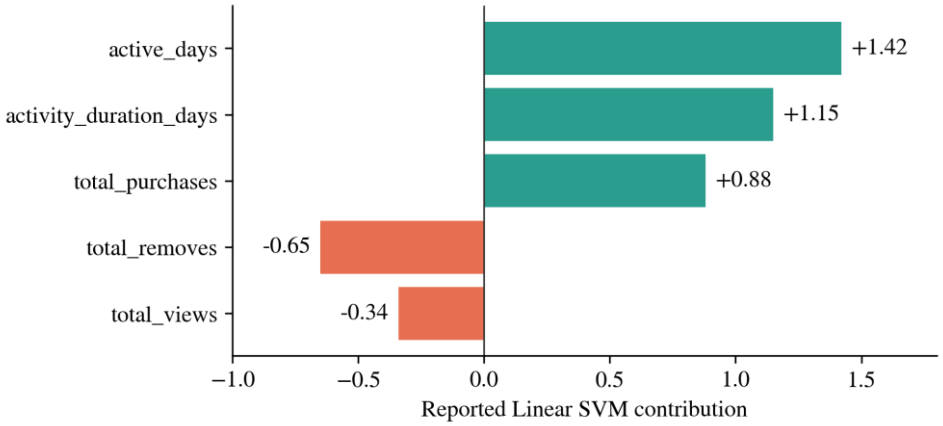


Figure 8: Reported Linear SVM feature contributions

4.1.6 Conclusion of Classification Analysis

The choice of model depends on the intended trade-off between false positives and missed retained users. Linear SVM offers the strongest F1-score of the tested models, but its precision remains modest. The output is therefore best used to prioritise further review or experimentation rather than to make automatic customer-level decisions.

4.2 Behavioural Cohort Segmentation Using K-Means

4.2.1 Introduction

Clustering groups users with similar behavioural patterns without using the target label to define the groups. ProductPulse applies K-Means to create behavioural segments that complement the supervised prediction.

4.2.2 General Description

K-Means [6] partitions the observations into K groups by assigning each observation to its nearest cluster centroid and iteratively updating the centroids until the assignment is stable. Because K-Means relies on Euclidean distance, the behavioural features are standardised first (Eq. (1)). The number of clusters is chosen by sweeping K and comparing silhouette scores.

4.2.3 Specific Requirements, Functions and Formulas

K-Means minimises the within-cluster sum of squares in Eq. (12), where S_k is the set of points in cluster k and μ_k its centroid. The quality of a clustering is assessed with the silhouette coefficient [7] in Eq. (13), where a(i) is the mean distance from observation i to the other members of its own cluster and b(i) is the smallest mean distance from i to the members of any other cluster. Values close to 1 indicate well-separated clusters.

$$J = \sum_{k=1}^{K} \sum_{x \in S_k} \|x - \mu_k\|^2 \tag{12}$$

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{13}$$

4.2.4 Analysis Results

The parameter sweep gives its maximum silhouette score of 0.8174 at K = 2, so two segments are used. Table 14 profiles them.

Table 14: Behavioural cluster profile

Measure	Cluster 0: Casual Explorer	Cluster 1: Highly Engaged User
Share of users	94.41%	5.59%
User count	258,093	15,273
Retention rate	7.40%	39.36%
Mean sessions	1.69	11.53
Mean purchases	0.17	7.85
Mean views	2.86	36.02

The Casual Explorer segment contains most users and has low engagement, with a retention rate of 7.40%. The Highly Engaged User segment is much smaller but has substantially higher sessions, purchases and views, and a retention rate of 39.36%, about 5.3 times that of the larger segment. The two cluster sizes sum to the full cohort of 273,366 users.

4.2.5 Visualization

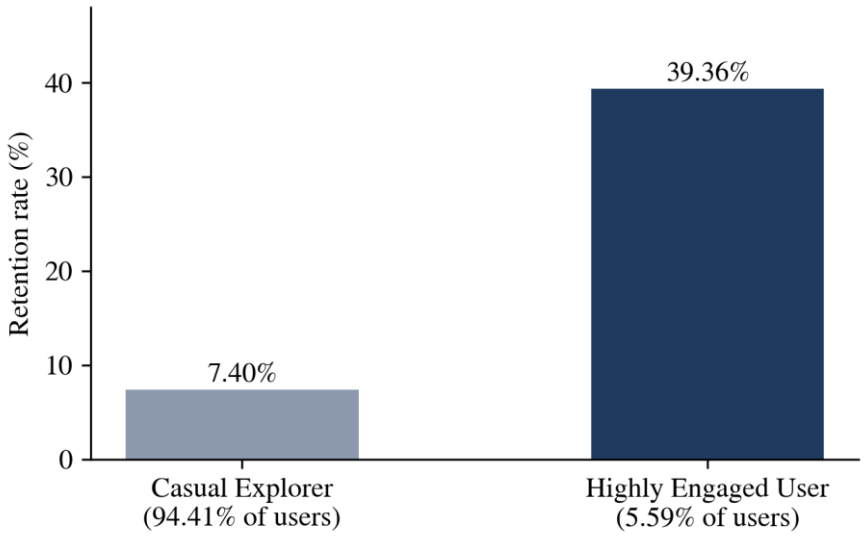


Figure 9: Retention rate by behavioural segment

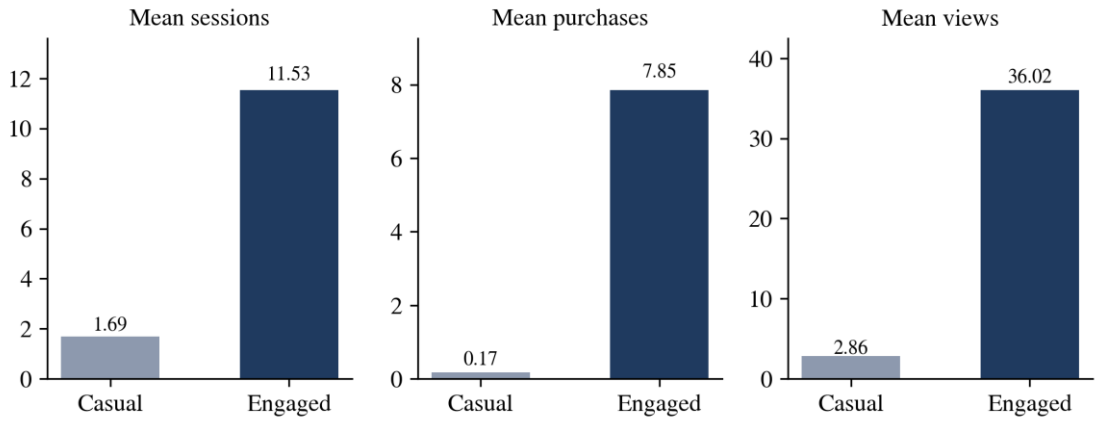


Figure 10: Mean sessions, purchases and views by segment

4.2.6 Conclusion of Clustering Analysis

The high silhouette score shows clear separation under the selected feature representation, but it should be interpreted with care: with a very large majority cluster and a small, heavy-tailed group of intensive users, a two-cluster solution separates the extreme users from everyone else almost by construction. The segments are interpretable and useful, for example for studying onboarding of low-engagement users or repeat-purchase patterns of highly engaged users, but any such action is a business hypothesis and not a tested intervention outcome of this project.

4.3 Dimensionality Reduction Using PCA

4.3.1 Introduction

Principal Component Analysis (PCA) [8] transforms correlated numerical variables into a smaller set of uncorrelated components that preserve as much variance as possible. In ProductPulse it provides a two-dimensional map of the behavioural feature space.

4.3.2 General Description

PCA is applied after centring and scaling the features. It computes principal directions from the covariance structure of the data. The first principal component captures the greatest possible variance, and each following component captures the greatest remaining variance subject to being orthogonal to the previous ones.

4.3.3 Specific Requirements, Functions and Formulas

For the standardised data matrix X with N observations, the covariance matrix is given by Eq. (14). The principal components are its eigenvectors, obtained from Eq. (15) and ordered by

descending eigenvalue λ_i. The share of variance explained by component i is given by Eq. (16).

$$\Sigma = \frac{1}{N-1} X^T X \tag{14}$$

$$\Sigma v_i = \lambda_i v_i \tag{15}$$

$$\mathrm{EVR}_i = \frac{\lambda_i}{\sum_j \lambda_j} \tag{16}$$

4.3.4 Analysis Results

Table 15: PCA explained-variance summary

Component	Explained variance	Interpretation
PC1	36.03%	Overall engagement volume
PC2	18.18%	Conversion efficiency
Cumulative (PC1 + PC2)	**54.21%**	—

The first two components together explain 54.21% of the total variance (36.03% + 18.18%). PC1 is interpreted as overall engagement volume and PC2 as conversion efficiency. The remaining 45.79% of the variance lies in further components, so the two-dimensional map is informative but not complete.

4.3.5 Visualization

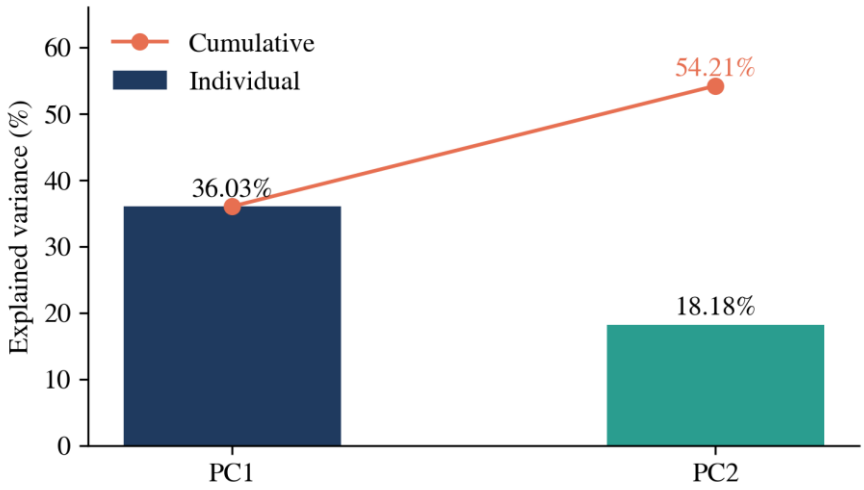


Figure 11: Variance explained by the first two principal components

The application additionally draws the users in the (PC1, PC2) plane, coloured by segment, together with the position of an analysed user; see the Segments screen in Section 5.2.

4.3.6 Conclusion of PCA Analysis

PCA gives a compact view of the behavioural data that is useful for communicating broad patterns. Any scatter plot in the PCA plane should be labelled as a projection, because distances and overlaps in two dimensions may not fully represent the original high-dimensional structure.

4.4 Sensitivity Analysis and Decision-Boundary Tipping Point

4.4.1 Introduction

The project includes a sensitivity simulation that shows how changes in selected behavioural inputs alter the Linear SVM decision score. It makes the model easier to understand and answers a practical what-if question: how much would a user's behaviour have to change before the model classifies the user as retained rather than at-risk?

4.4.2 General Description

The Linear SVM decision boundary is the hyperplane f(x) = 0. Users with f(x) < 0 are classified as at-risk and users with f(x) ≥ 0 as retained. Because f is linear, the effect of a change in one feature on the score is constant and can be computed in closed form, which allows the simulation to run instantly in the interface.

4.4.3 Specific Requirements, Functions and Formulas

Since the model operates on standardised features, a one-unit increase in the raw feature x_j changes the score by the marginal gradient in Eq. (17), where w_j is the SVM weight and σ_j the standard deviation of the feature. The smallest change in feature j, with all other inputs held fixed, that moves an at-risk user (f(x) < 0) onto the boundary is given by Eq. (18).

$$\frac{\partial f(x)}{\partial x_j} = \frac{w_j}{\sigma_j} \tag{17}$$

$$\Delta x_j^* = \frac{-f(x) \cdot \sigma_j}{w_j} \tag{18}$$

Features with large |w_j|/σ_j reach the boundary with small behavioural changes. In practice several features change together (for example, one extra purchase also changes total_events and purchase_view_ratio), so the combined effect can differ from the single-coefficient estimate.

4.4.4 Analysis Results

In the reported simulation, incrementing purchases produces an approximately 2.1 times higher boundary-crossing velocity than incrementing passive sessions. In one example, an abandoned-cart user with a decision score of f(x) = −0.76 moves to f(x) = +0.18, which crosses the boundary, after a single completed purchase is added.

4.4.5 Visualization

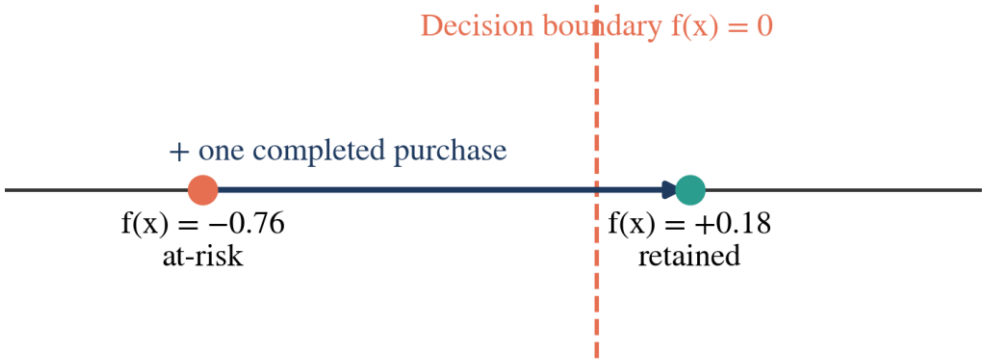


Figure 12: Reported tipping-point example: one purchase moves the score across the boundary

4.4.6 Conclusion of Scenario Analysis

Sensitivity analysis helps explain which inputs most strongly affect the model score and supports what-if exploration. It is a model-based counterfactual: it does not prove that causing a customer to make one additional purchase would produce the same real-world retention outcome. The interface should therefore describe the result as a simulated score change rather than a guaranteed improvement.

4.5 Illustrative Retention Economics

4.5.1 Introduction

This analysis translates hypothetical recovery-rate scenarios into indicative gross merchandise value (GMV) and customer acquisition cost (CAC) avoidance. It is an illustrative calculation only; the dataset does not provide verified AOV or CAC values.

4.5.2 General Description

For a given assumed recovery rate L (in percent) among the at-risk users, the number of users preserved is N_at-risk × L/100. Each preserved user is assumed to place one order of average value AOV and to avoid one acquisition cost CAC. The scenarios of 1%, 5% and 10% are used to show how sensitive the totals are to the assumed lift.

4.5.3 Specific Requirements, Functions and Formulas

Table 16: Inputs of the economic illustration

Input	Assumed value	Status
At-risk user base (N_at-risk)	248,253	Project-reported cohort count
Average order value (AOV)	₹2,500	Hypothetical assumption
Customer acquisition cost (CAC)	₹800	Hypothetical assumption
Recovery-rate scenarios (L)	1%, 5%, 10%	Illustrative scenarios

$$R_{\text{recov}}(L) = N_{\text{at-risk}} \times \frac{L}{100} \times \text{AOV} \tag{19}$$

$$S_{\text{CAC}}(L) = N_{\text{at-risk}} \times \frac{L}{100} \times \text{CAC} \tag{20}$$

$$V(L) = R_{\text{recov}}(L) + S_{\text{CAC}}(L) \tag{21}$$

The combined value V(L) is not profit, net present value or ROI. It adds two different illustrative quantities and does not account for campaign cost, margin, repeat purchases, timing, or overlap between revenue and acquisition economics.

4.5.4 Analysis Results

Table 17: Illustrative retention-value scenarios (1 Crore = 10 million rupees)

Recovery rate	Recovered users	Indicative GMV	Indicative CAC avoidance	Combined value
1%	2,483	₹0.62 Cr	₹0.20 Cr	₹0.82 Cr
5%	12,413	₹3.10 Cr	₹0.99 Cr	₹4.10 Cr
10%	24,825	₹6.21 Cr	₹1.99 Cr	₹8.19 Cr

The values scale linearly with the assumed recovery rate: doubling the recovery rate doubles every column. They remain illustrative until AOV, CAC, the eligible population and a defensible recovery-rate estimate are validated.

4.5.5 Visualization

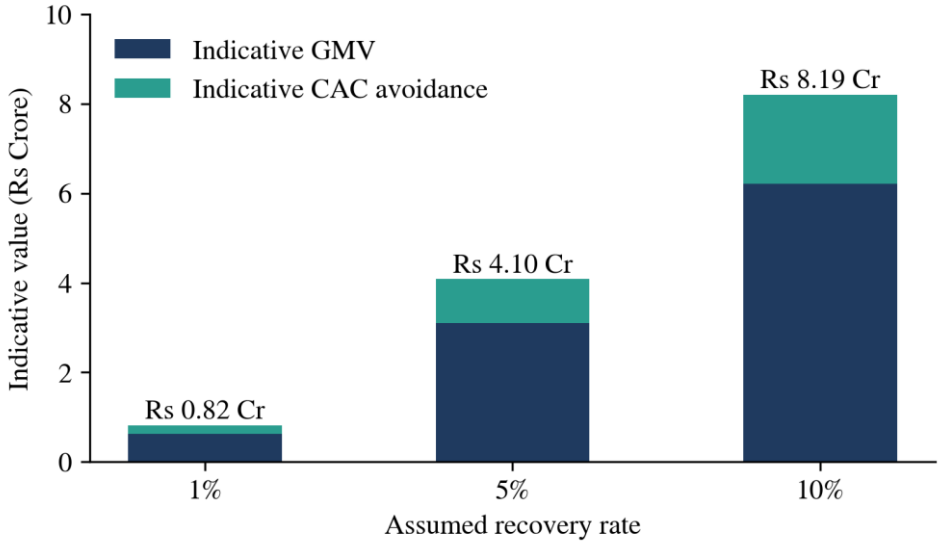


Figure 13: Illustrative value by assumed recovery rate (Rs = Indian rupees)

4.5.6 Conclusion of Economic Illustration

The scenario table demonstrates how retention assumptions can be translated into indicative financial magnitudes. It is not evidence that ProductPulse has generated the stated revenue or savings, and it should not be presented as realised ROI.

5. CONCLUSION

5.1 Synthesis of Findings

ProductPulse demonstrates a workflow for converting raw e-commerce interaction events into user-level behavioural features and using those features for short-horizon inactivity prediction, behavioural segmentation and exploratory dimensionality reduction. The classification comparison shows that Linear SVM has the strongest F1-score (0.3586) among the four evaluated models, with a ROC-AUC (0.7430) that is effectively tied with the others, while the class imbalance makes accuracy alone insufficient for judging model usefulness.

K-Means produces two interpretable behavioural cohorts with clearly different engagement and retention profiles (7.40% versus 39.36% retention). PCA reduces the feature space to two components that explain 54.21% of the variance. The sensitivity and economic sections provide ways to explore model behaviour and possible business value, but neither establishes causal impact or realised financial performance.

Table 18: Key results at a glance

Analysis	Main result
Dataset	273,366 labelled users; 9.19% retained, 90.81% churned / at-risk
Classification	Linear SVM selected: F1 = 0.3586, precision = 0.2829, recall = 0.4895, ROC-AUC = 0.7430
Segmentation	K = 2 (silhouette 0.8174): Casual Explorer 94.41% of users (7.40% retained); Highly Engaged 5.59% (39.36% retained)
PCA	PC1 + PC2 explain 54.21% of the variance (36.03% + 18.18%)
Sensitivity	One additional purchase moves the example score from −0.76 to +0.18
Economics (illustrative)	5% recovery of 248,253 at-risk users → 12,413 users, ₹4.10 Cr combined indicative value

5.2 Interactive Web Platform

The analyses are delivered through the ProductPulse application, which brings together user-level prediction, behavioural segmentation, model performance and retention-oriented scenario exploration. It uses a Python/FastAPI backend and a React/TypeScript frontend. Its main areas are:

- **Dashboard:** a high-level summary of the dataset and analytical outputs.

- **Analyze User:** accepts behavioural inputs and displays the model's prediction and decision score for an individual user profile.
- **Segments:** presents the behavioural cohorts, their profile summaries and the PCA map.
- **Performance:** presents model evaluation metrics for comparison.
- **Insights / Scenarios:** communicates behavioural findings, the sensitivity simulation and the retention-economics calculator.

The following figures show the screens of the running application.

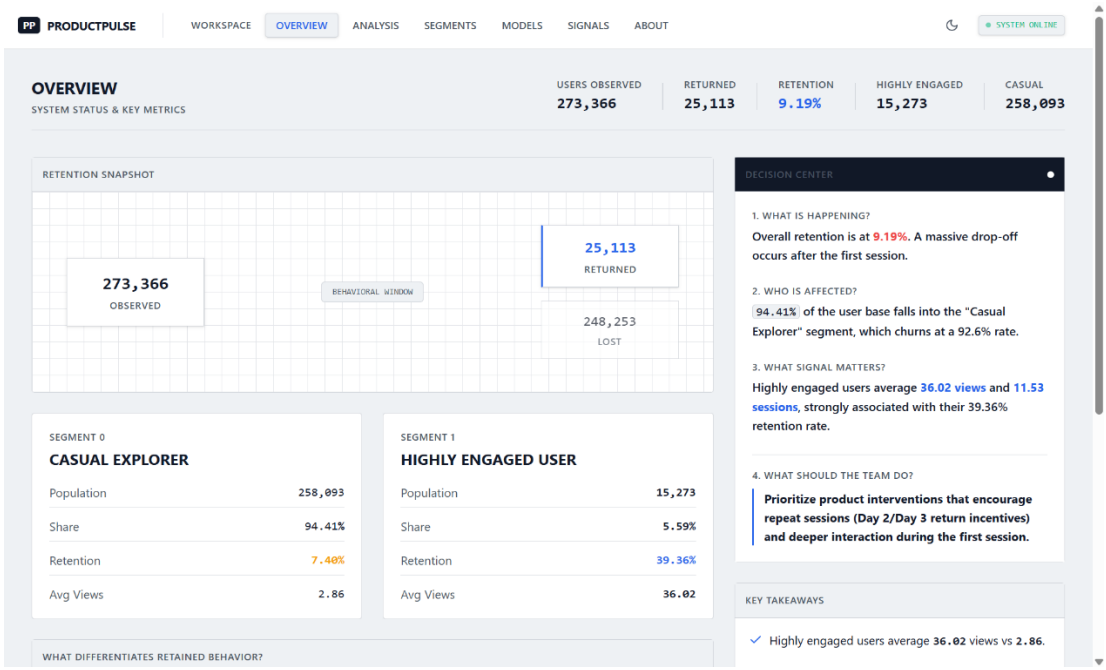


Figure 14: ProductPulse main dashboard

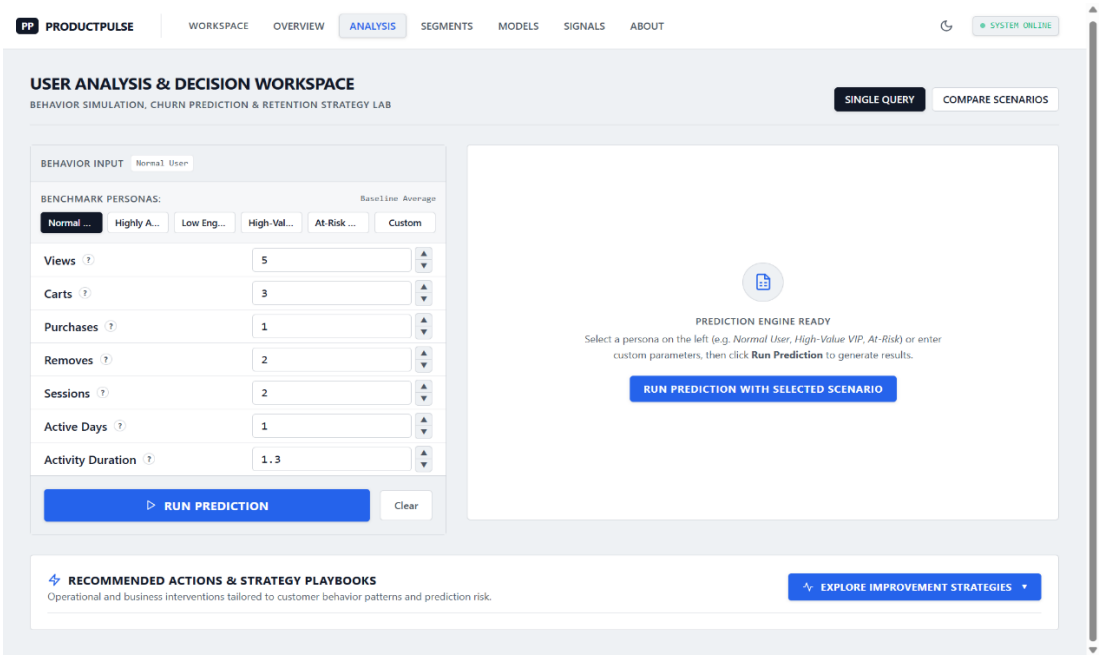


Figure 15: Individual user analysis screen

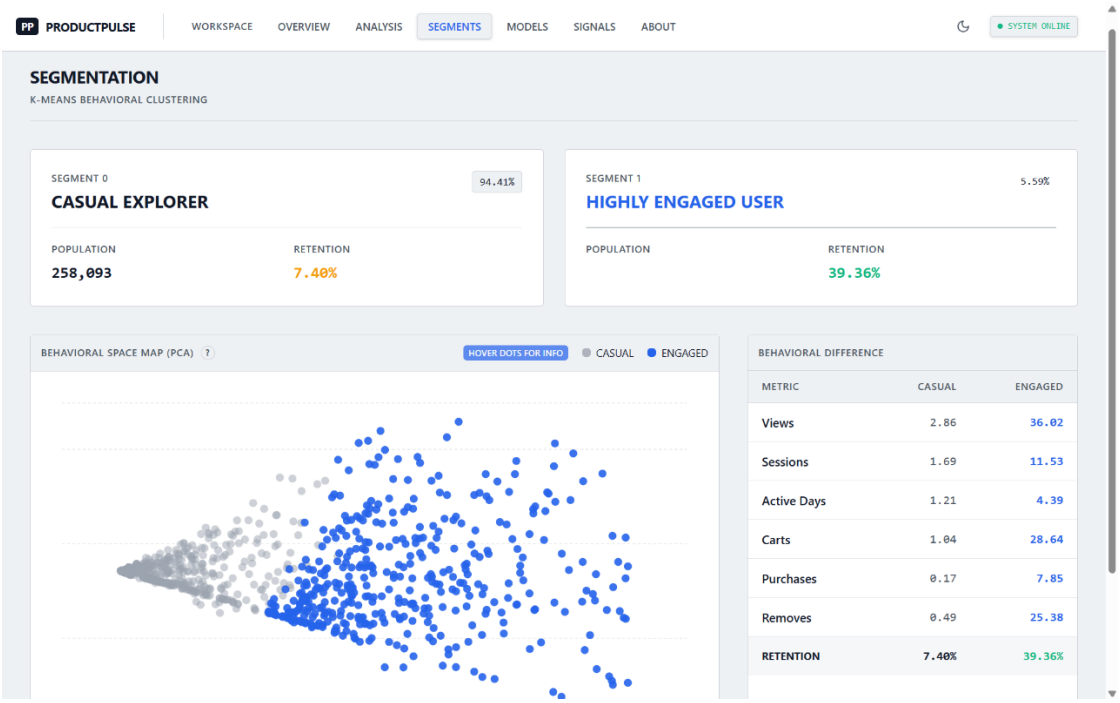


Figure 16: Behavioural segments screen

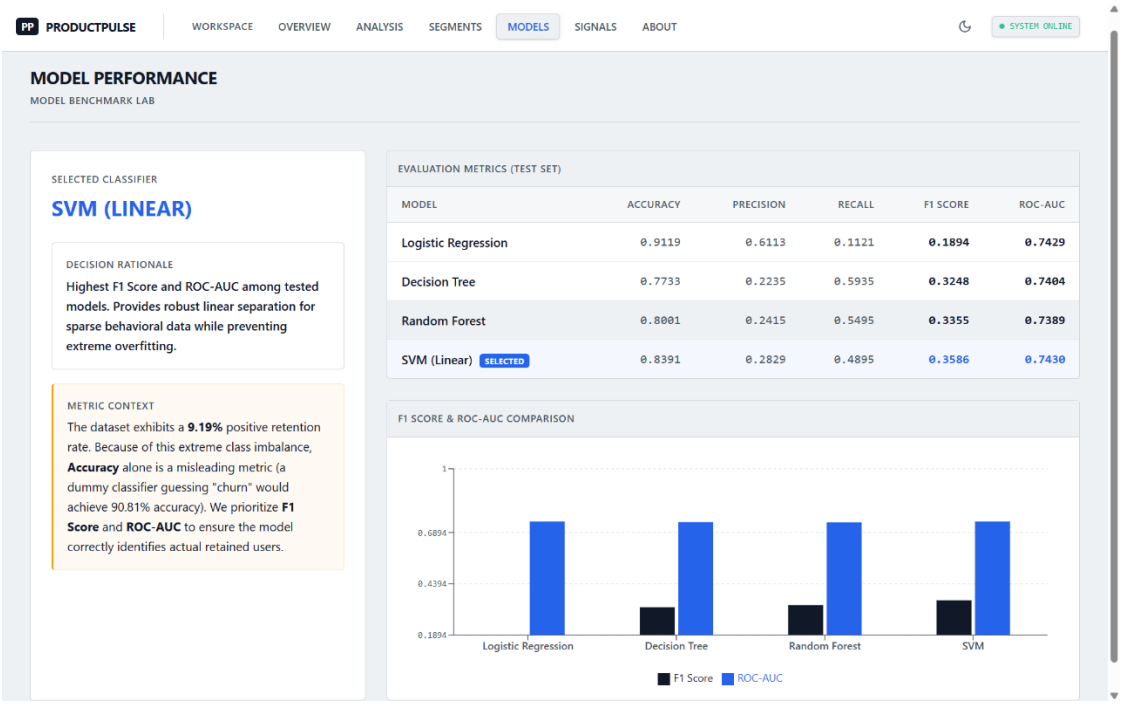


Figure 17: Model performance screen

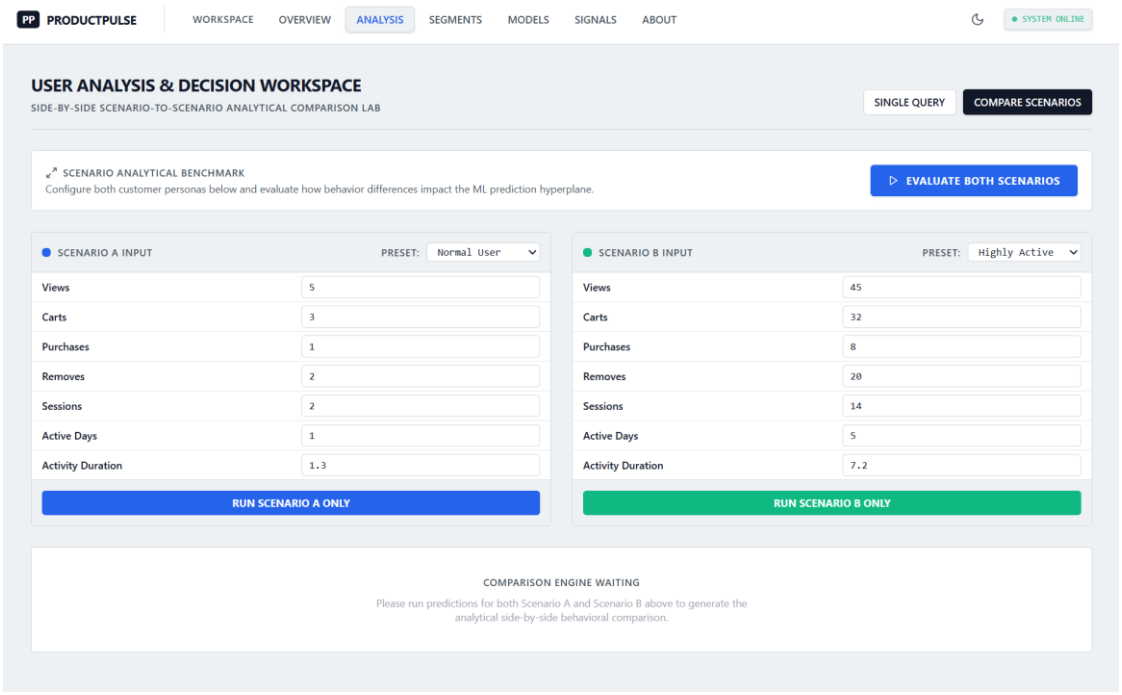


Figure 18: Scenario comparison and retention economics screen

5.3 Limitations and Recommendations

The following limitations should be kept in mind when the results are used:

- **Target definition.** The label measures inactivity in a ten-day window, not permanent churn. The windows are unequal in length (21 and 10 days), and the target window includes the Christmas period, which may affect activity patterns.
- **Single month.** All results come from one monthly file; stability across months has not been tested.
- **Modest precision.** The selected model has an F1-score of 0.3586 and a precision of 0.2829, so predictions are best used for prioritisation.
- **Model comparison.** The ROC-AUC values of the four models are almost identical, so the choice rests on the F1/recall trade-off at the chosen operating point.
- **Features.** No recency feature (days since the last event) is included, although recency is often a strong signal of disengagement. Correlated count features limit the interpretability of individual coefficients.
- **Clustering.** The very high silhouette score is partly a consequence of the skewed, heavy-tailed data and the dominant majority cluster.
- **Economics.** AOV, CAC and recovery rates are assumptions, and the combined value is not a profit or ROI estimate.

6. FUTURE SCOPE

The project can be extended in the following directions:

- **Temporal validation:** evaluate the model across multiple months and rolling time windows to test stability and reduce dependence on a single month.
- **Improved target definition:** distinguish temporary inactivity, repeat-purchase delay and verified customer churn where suitable labels are available.
- **Recency and sequence features:** add days since last activity, session gaps and trend features to the user representation.
- **Class-imbalance handling:** investigate class weights, calibrated probabilities, threshold tuning and carefully validated resampling.
- **Model explainability:** add local explanations and feature-contribution views, clearly distinguishing association from causation.
- **Advanced sequence modelling:** explore recurrent, transformer-based and graph neural network methods if event histories and sufficient computational resources are available.
- **Streaming analytics:** investigate event-stream processing tools such as Apache Kafka for near-real-time feature updates and scoring.
- **Business-system integration:** connect validated predictions to customer data platforms or marketing automation tools, subject to privacy, consent and governance requirements.
- **Controlled experimentation:** test retention interventions through A/B tests or other experimental designs before claiming incremental impact.
- **Monitoring and governance:** monitor drift, performance by segment, fairness, data quality and prediction feedback after deployment.

7. REFERENCES

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